

Tracing the dust and ISM properties of distant quiescent galaxies with ALMA and JWST

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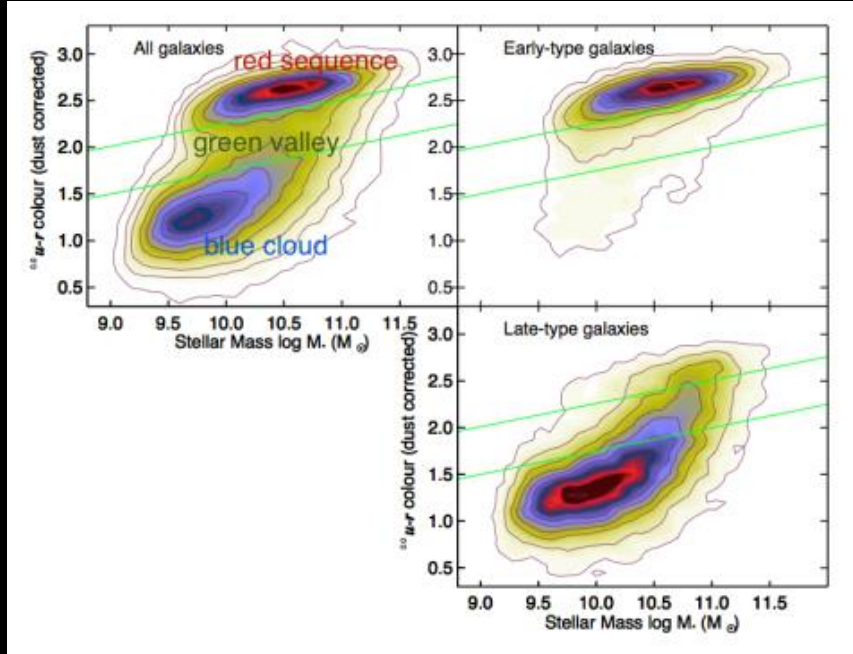
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DAWN

Quenching 101

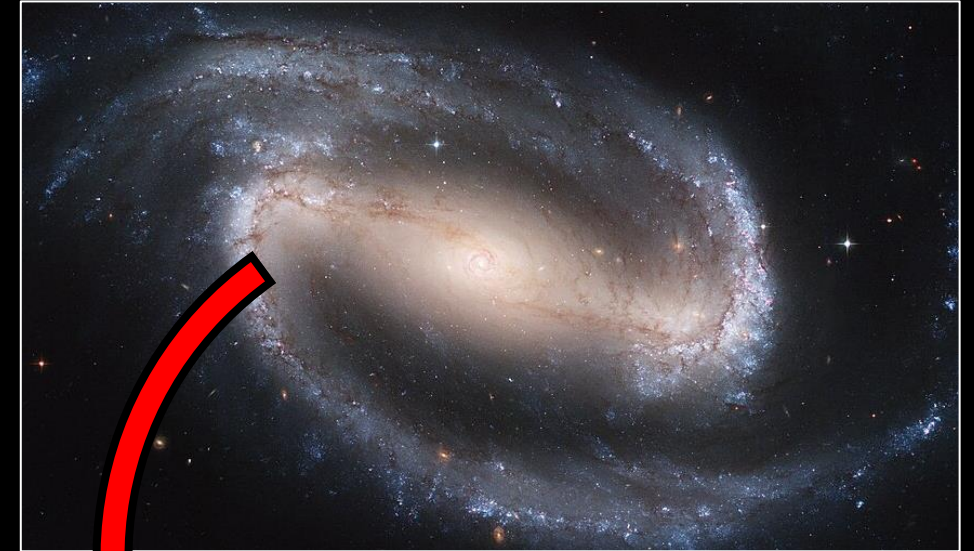
- Data from observational surveys tells us that the overall population of galaxies is **bimodal**



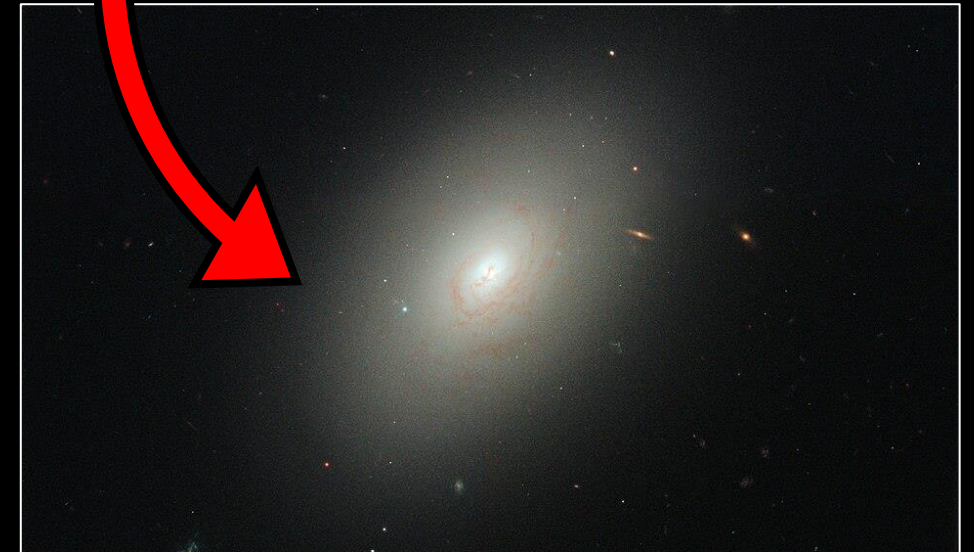
- Distinction between 1) **blue**, *late-type* (disk-like), star-forming galaxies and 2) **red**, often *early-type* (elliptical/not a disk), passive galaxies
- **Quenching**: The stopping* of star formation in a galaxy

* Not exactly a dead stop, just reduced drastically

Hubble image of NGC 1300, a star-forming spiral galaxy (Credit: NASA/ESA)

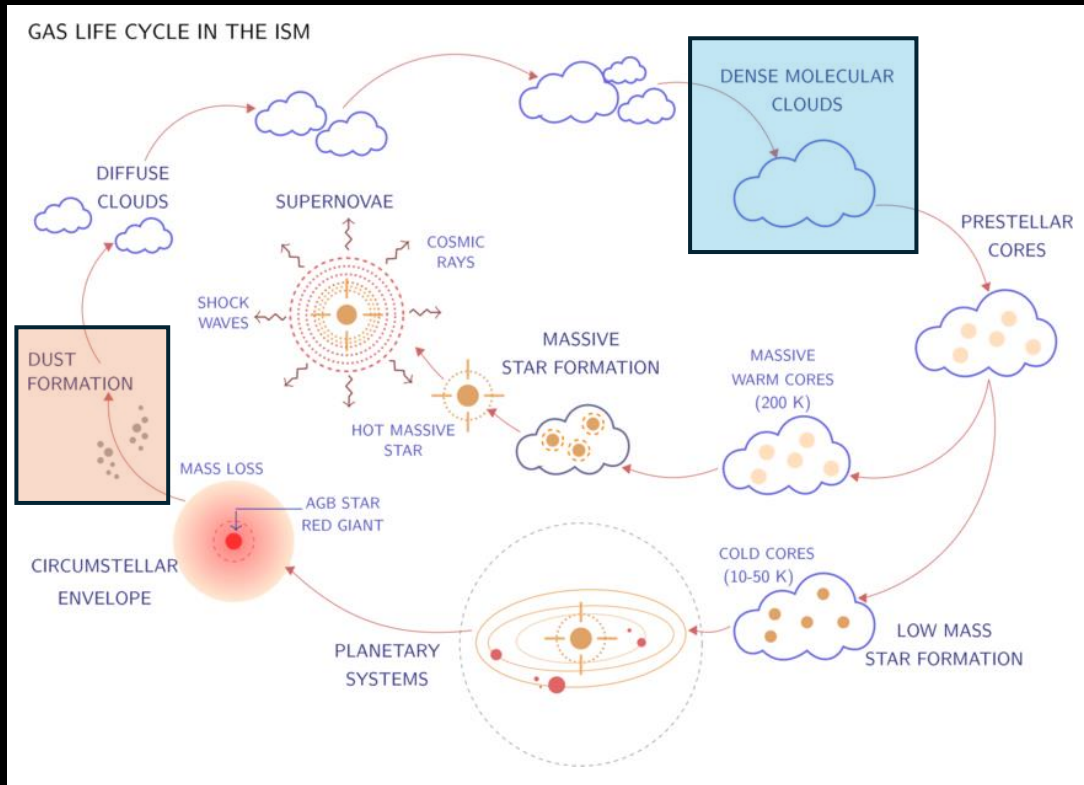


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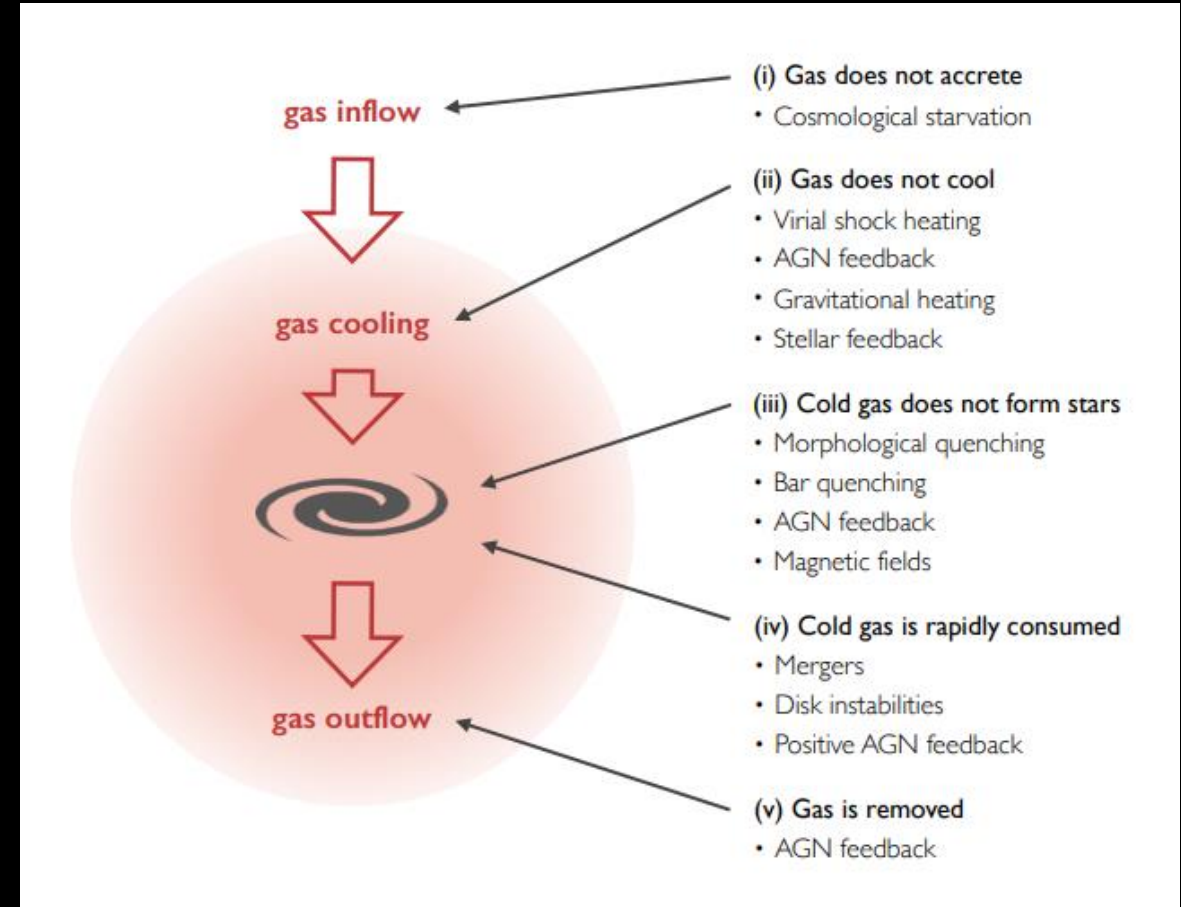
Hubble image of NGC 4150, a quiescent elliptical galaxy (Credit: NASA/ESA)

Why quenching happens/the life cycle of stars



- Stars need fuel to form: **Cold gas** and **dust**
 - Cold gas** is made up of hydrogen and helium which are ingredients for fusion
 - Dust** enables cooling of gases by absorbing energy, so thermal pressure doesn't counteract gravity pressure

- Quenching can be entanglement of several gas starving/heating/exhausting processes

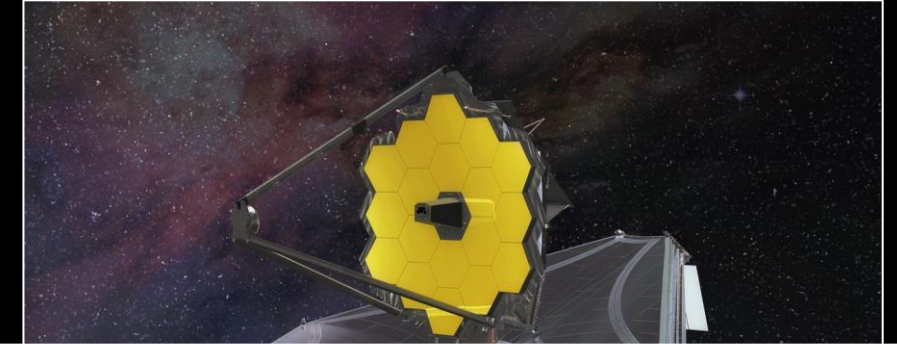


Schematic of quenching mechanisms (Man and Belli, 2018)

What JWST has to say

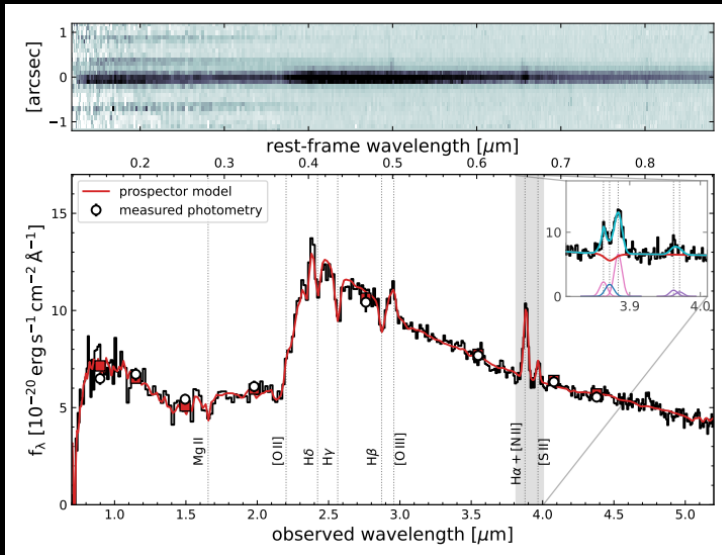
(Credit: NASA/ESA)

- *James Webb* has unprecedented sensitivity in infrared
 - Convenient for observing features in rest-frame UV, visible wavelength range emitted by high-redshift galaxies
- What did it find: An **overabundance** of quiescent galaxies (QGs) who formed their stars and stopped within 0.3-3 billion years after the Big Bang:



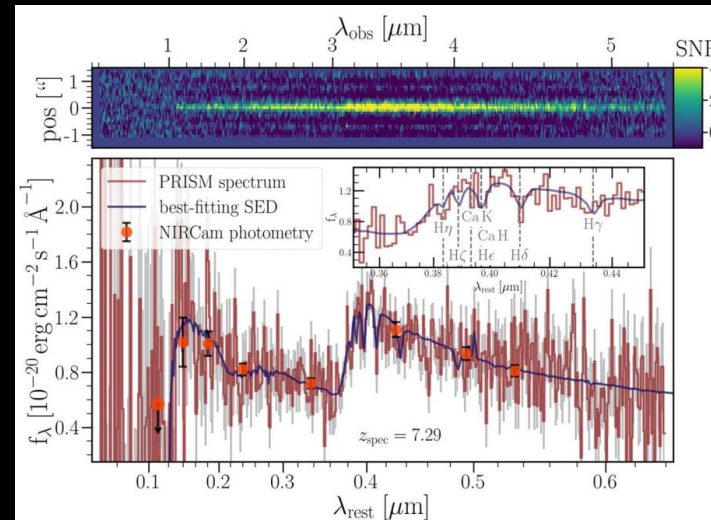
Question: If galaxies can form and die much earlier in cosmic history than expected, **what are we missing?**

A QG at $z = 4.9$ (1.2 Gyr after Big Bang) that formed its stars in a 200Myr burst



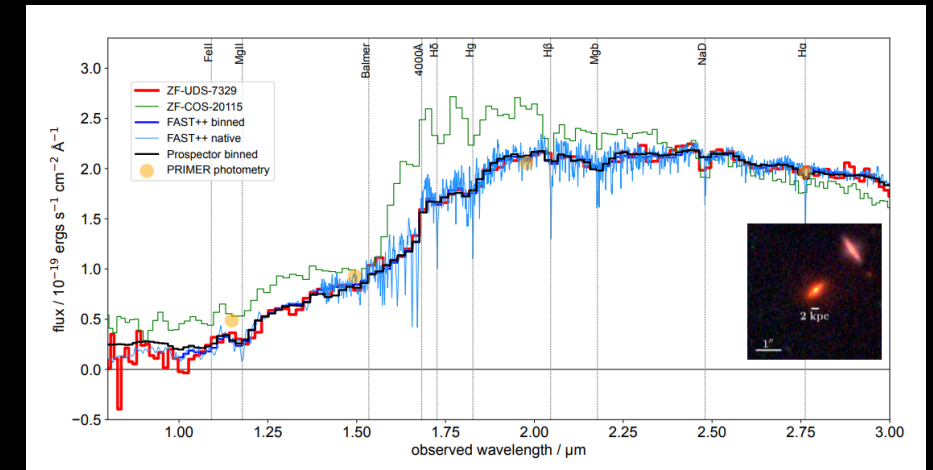
(de Graaff et al. 2025)

A massive QG at $z = 7.3$ (700 Myr after Big Bang) that ceased forming stars 100Myr prior



(Weibel et al. 2025)

A massive galaxy at $z = 3.2$ that formed its stars by $z = 11$ (400 Myr after Big Bang)



(Glazebrook et al. 2024)

What am I doing about this?

- To examine quenching at high-redshift, we must examine the **fuel** remaining
- My project looks at investigating this with two different approaches on two different sample sets, to broadly examine QG properties:

How much dust is retained in QGs?

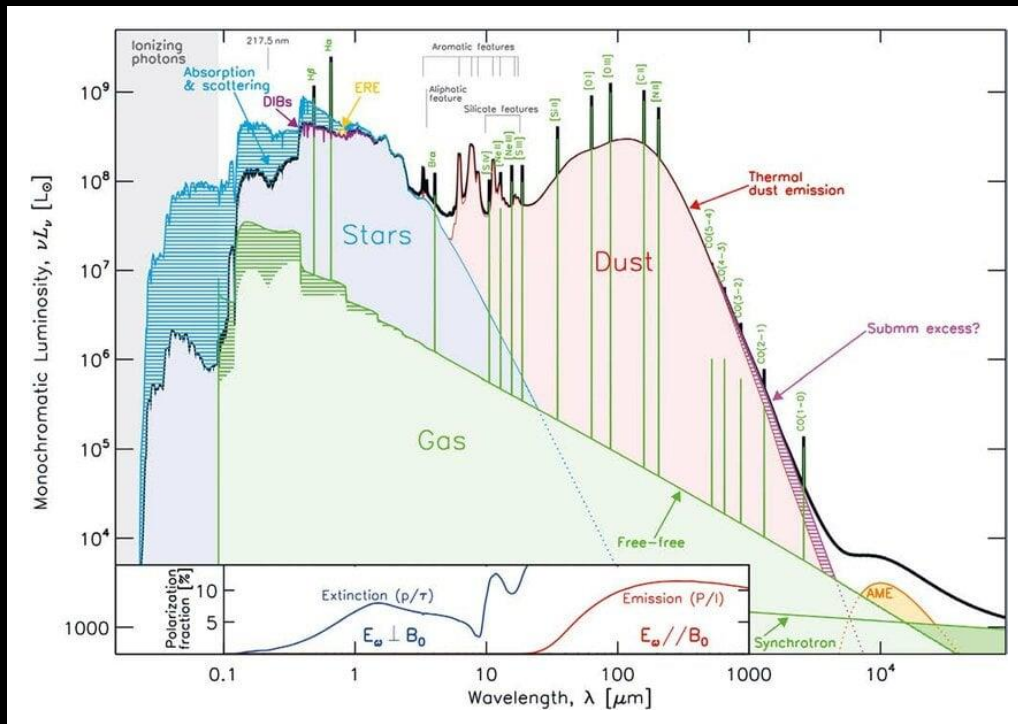
What are the ISM conditions of QGs?



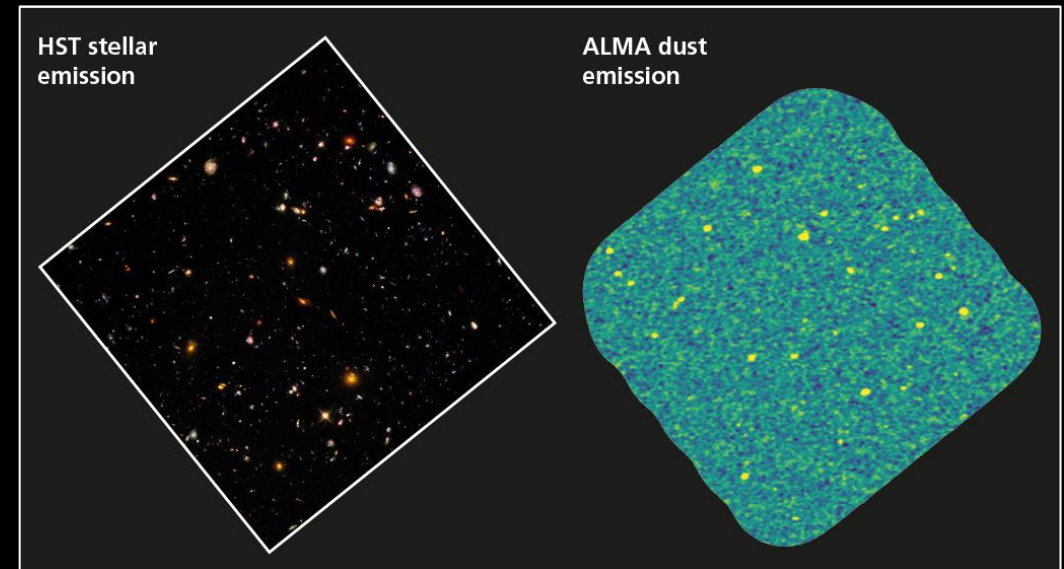
Observing dust with ALMA

- ALMA (Atacama Large Millimetre/submillimetre Array) is a telescope array that observes radiation from cold dust and gas
 - Utilises interferometry (combining signals from multiple dishes)
 - Dust absorbs UV light from bright young stars, re-emitting it as **far-infrared radiation** (approx. 8-1000 μ m)
 - ALMA observes between 0.3-3.6mm, so high-z dust is observable

Credit: Alma Observatory



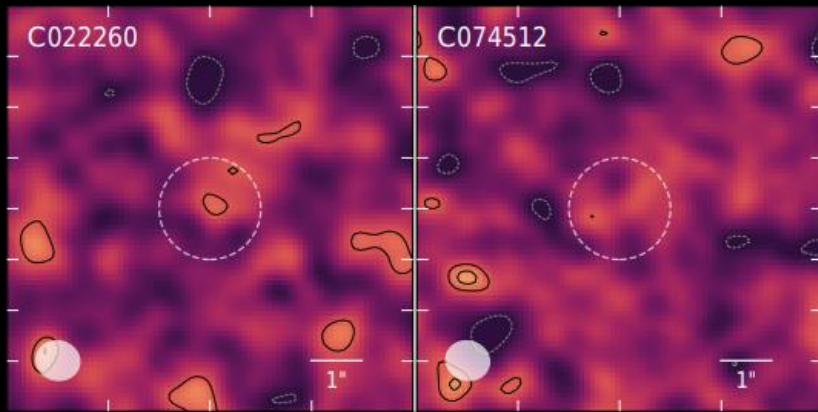
(Sauvage, 2020)



ASPECS fields. Credit: ALMA Observatory

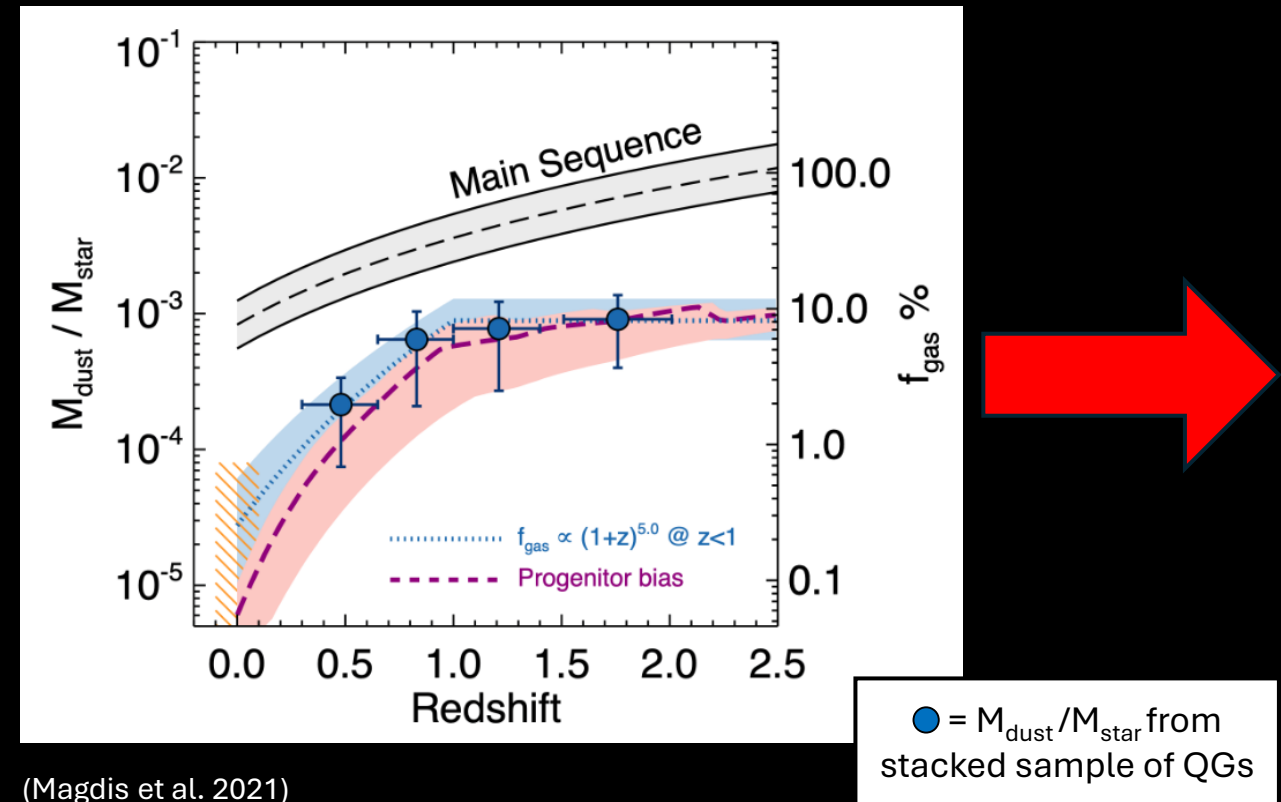
How much dust is there in high-z QGs

- Dust measurements exist for quiescent galaxies observed at $z = 0$ to $z \sim 2.5$
- My aim: Look at dust at **higher redshifts**
- Problem: Sources dim $\sim 1/d^2$
- What we can see with ALMA at high- z :
 - Dusty star-forming galaxies (lots of bright dust!)
 - Strongly lensed sources (magnifies the source)
- Quiescent galaxies = Not star-forming, so not dust producing, so should be faint in dust



- Solution: Stack lots of sources together!

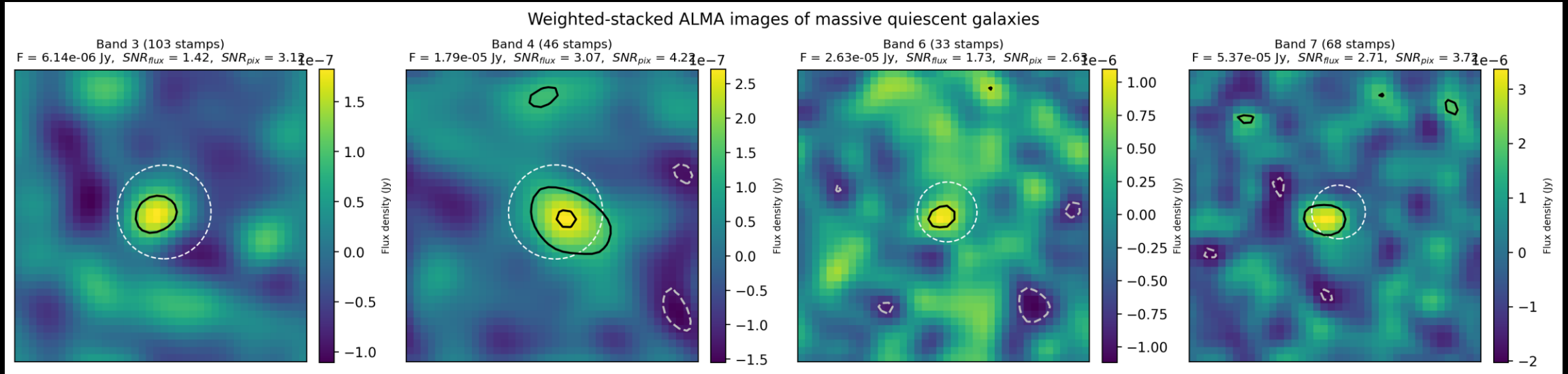
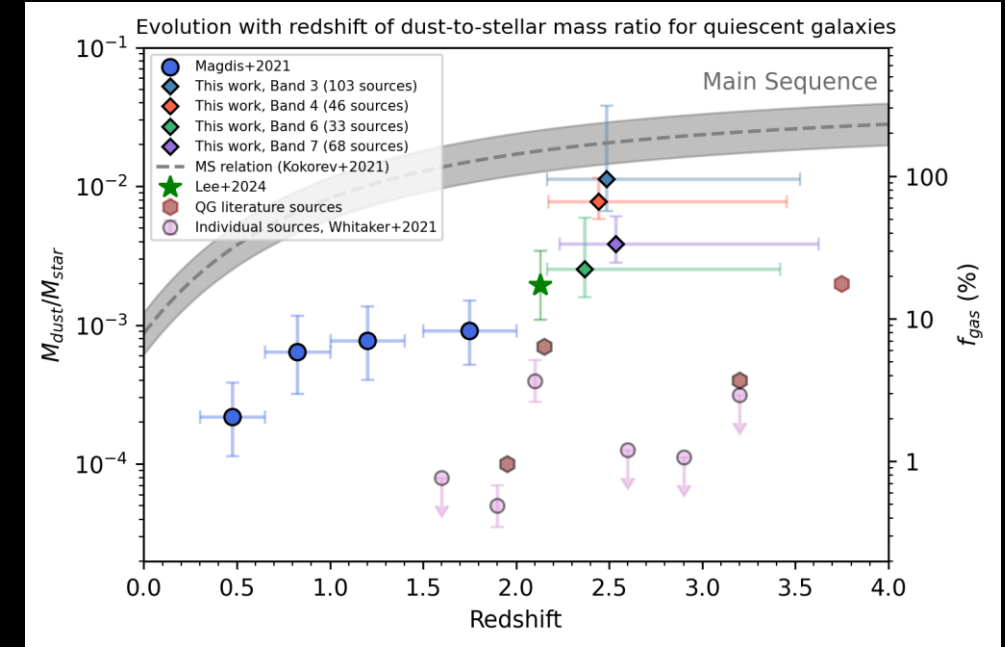
Dust-to-stellar-mass ratio of QGs vs redshift



(Magdis et al. 2021)

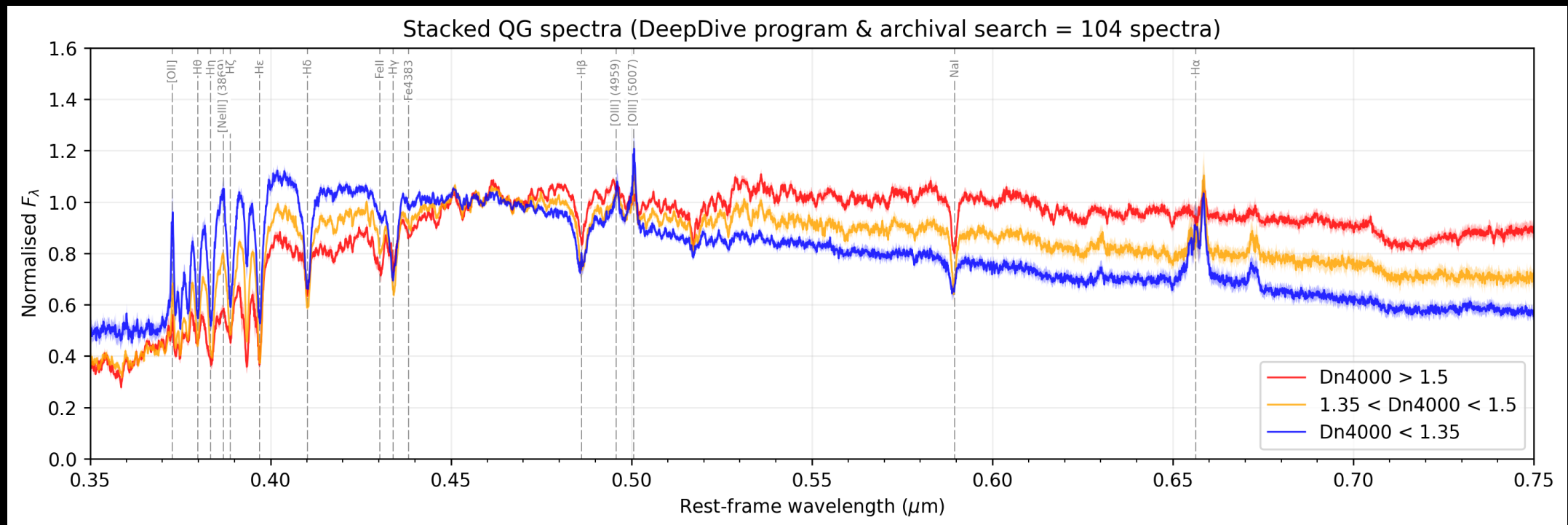
Stacking sources and measuring dust

- Stacking observations together and averaging out per-pixel increases signal-to-noise ratio and allows for clearer dust detection
 - Often segmented by physical parameters: time since quenching, stellar mass, redshift
- 250 cutouts sourced from existing ALMA data (where available) for QGs confirmed by JWST/NIRCam imaging
 - Stacked together by ALMA band, different observed wavelength range
 - Sources range between $z = 2$ and $z = 7$, median ~ 2.5 for each band
- Results: Massive QGs might have large dust reserves, but not a confident result (flux SNR of 1.4–3.1 for each band)



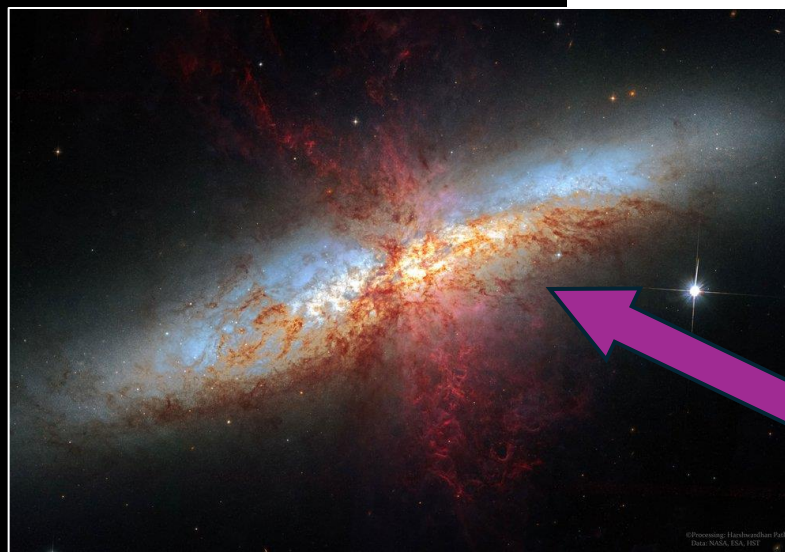
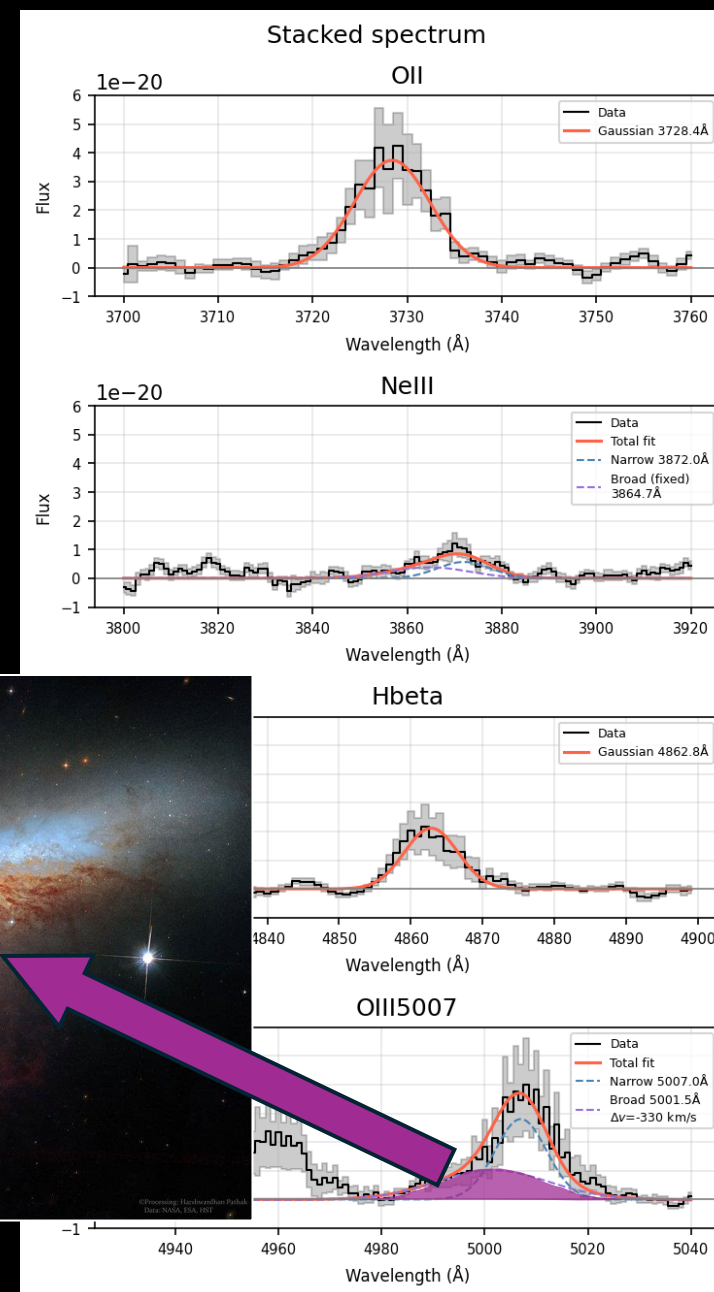
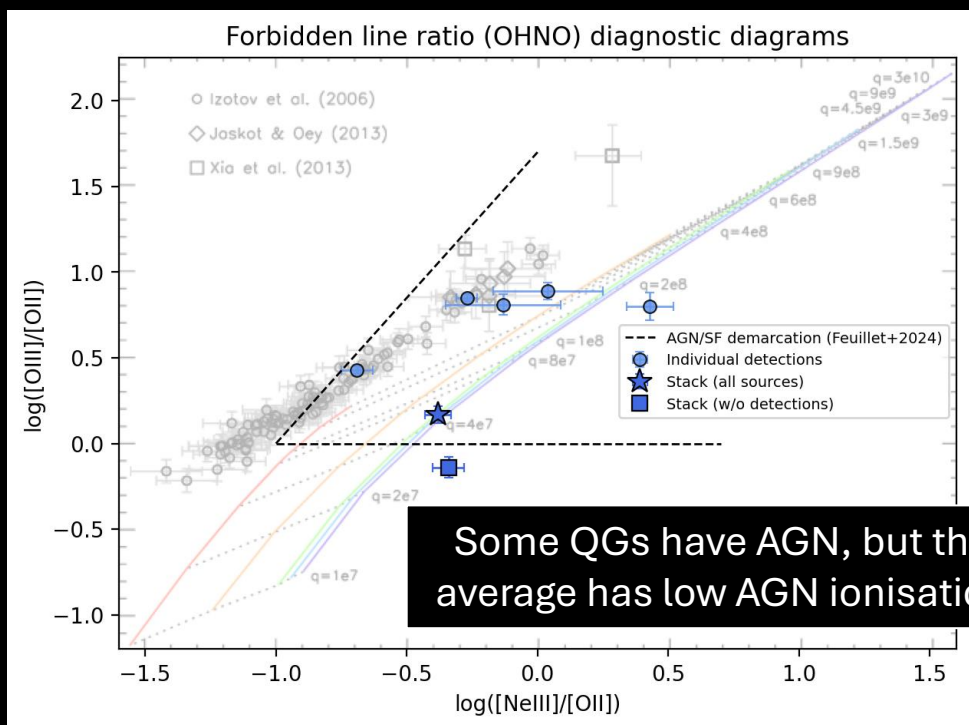
Examining Gas/ISM properties from JWST spectra

- Spectral line features observed in galaxy spectra can tell us about the structure, age, and gas contents
- Gas being expelled/heated by feedback from the energetic active galactic nucleus (AGN) can be traced through redshifting of certain emission lines
- By examining lines that can only result from AGN (where high ionisation potentials can be reached), we can quantify whether these galaxies are quenched because of AGN, or despite them



Line ratios to probe AGN strength

- Using 104 JWST/NIRSpec spectra, we segment them by galaxy age, and stack them together weighted by signal-to-noise
 - Favours less noisy spectra with clear lines
- We measure the ratio of emission line fluxes in individual galaxies and the stacked sample, which tells us what's ionising the gas
 - AGN produce harder radiation fields than stars, so [OIII] and [NeIII] (higher ionisation energies) are useful to trace AGN activity – and **outflows** !



Summary

- Several processes lead to the quenching of galaxies, exhausting their fuel reserves
- Observing quiescent galaxies in the early universe can challenge our cosmological models – Thanks, JWST!
- Constraining dust masses through stacking tells us that high-redshift QGs are still dust-rich (maybe?)
- Examining emission lines tells us that AGN (evidenced by line ratios and outflows) might be responsible for quenching in some high- z QGs (maybe)

Questions?

